

ActInf Textbook Group ~ Cohort 2 ~ Meeting 1

(Welcome back, Coda & Learning Practices overview)

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hello everyone it's the active inference textbook group welcome to everyone who's joining it is cohort two's meeting one on september 2nd 2022 so welcome everybody um just to mention gather features and then let's just have a few minutes to say hello if we would like on the bottom bar is where you can mute yourself or turn off your video so that's helpful for people on the bottom bar also there's a smiley face where you can see a raised hand and the hand raise which you can also reach with just hitting six is how you have to lower it manually but that's how we'll know that people want to speak which is like super helpful and moving with the arrow keys but everyone's figured out if they're here the main asynchronous tool is going to be the coda but first before we go through the coda um let's just take a few minutes and anybody who wants to can just say hello if they want i want to ask nail to make a few steps from the entrance to make it open okay neil oh thank you all right nice does anybody want to just say anything anyone returning from um the previous cohort who would just want to say anything or anything else before we launch into more of like the overview of the coda someone can just speak up they don't need to raise their hand ali and then anyone else hi i'm ali i'm an independent researcher from iran this is the second time i'm joining the textbook uh the previous cohort was fascinating in my opinion and i hope to meet a lot more people on this cohort as well and looking forward to another fascinating journey together thanks thank you ellie anyone else can raise their hand or give any thought thank you i can say hi i'm blue i'm also joining from the first cohort and i will be um helping sometimes when daniel's not around so it's nice to meet all of you guys and i'm excited to get deeper into the machine all right awesome ragnar and then anyone else who wants to raise their hand yeah hey everyone that's my first cohort but really excited to be joining um i'm currently like a data scientist at the at q bio it's a startup in san francisco building like a digital twin of the human body but i've always been very fascinated by the active by active influence also read the new book but excited to be

joining the court and meeting other people who are passionate about the topic awesome yvonne and then anyone else who wants to raise your hand yeah hello my name is ivan i'm based in iran and in acting institute i direct a another education program and here i'm in a student role all right well this is the second cohort and there's some people who are also rather rejoining this first part of the book or and or in cohort two uh or in cohort one part two and the meetings are following so if you want to stick around or if you're in both that's totally cool i think we're obviously just learning by doing with even the structuring so hopefully people can start to think about active inference and how it applies to our processes and uh iterate and improve even within cohorts within these next several months together anyone can also write questions in the chat which is the second um [Music] set of button next to the people the first part of this onboarding is going to be in the main single source of truth which is posted in the nearby chat so this is the document to bookmark and use and contribute to as well as any other ways that you want to be contributing um like in the institute or just in your own work it's all good however you're being engaged and this is like our shared information environment everyone should have edit access but this is fine time to check we're gonna go through some of the pages and how they've been used in the previous cohort it's extremely rearrangeable so we we wanna and and people can make copies of things and we'll go through some features and so on and just talk about how we'll use these different pages and everything and if anybody like from the previous cohort or who has a question about something you can just raise your hand or use the chat and like there's it's just good to clarify things as we go so the first page the top page here is an overview of this textbook group and you can right click and download the textbook pdf itself it's an open source textbook it's the march 2022 active inference textbook with par pizzulo and fristen some people probably already completed scanning it over or reading it deeply others are probably going to go download it the directory page gives some information about what these different coda pages are participation rules and guidelines this is something to note and also provide your feedback on which is the synchronous and asynchronous aspects of this textbook group which is to say what we discuss in recorded and or active institute synchronous conversations and what we put into our shared knowledge repository unless it's signaled otherwise can be considered as having this creative commons attribution status which is citation open source with attribution so just kind of keep that in mind and as we explore um learning and design decentralized science together it will be important to have people's feedback but just wanted to note that because it's kind of an important aspect of doing work and it helps structure what kinds of contributions you want to

be making in this shared document where truly all contributions and questions are really important the most basic most speculative which we'll talk about but this is what the status of those contributions are and we're here as students and individuals who are learning and applying active inference especially in the second half of the book and of course probably in all of our thinking on active inference or even day-to-day work we're very interested in applying active inference also this is a textbook group that has a learning in a student basis but that's going to be happening in new ways inside and outside the group and so again something to kind of just consider and give feedback on so engagement will look different for different aspects even with planning we still may not know like what the synchronous and asynchronous availabilities are for your engagement with a textbook and with the group all the meetings are recorded which we'll talk about but mainly it's going to be about your asynchronous engagement with the textbook itself and some different practices around that anything else that you see as part of your learning journey here could be contacting people who you're connecting with participating in a discord getting involved in some other projects or activities that we have looking more at projects with other participants so it's not all just about reading and and um we'll try to unpack that you all have access to this coda document some pages are a little bit fixed and the ones especially with tables are there to be like expanded upon and we'll get through that we um have two ongoing cohorts so if people would like in the cohort too how we are here by your name um or let us know if you're not added we might just need to sync something you could note whether you've downloaded the textbook made time in your calendar for these synchronous meetings if it works or like making time to re-watch it or just making time to be paying attention to this area and then if um you're uh like wanting to share some blogging or potentially to do some writing around your work and if you want to place a link to your site and contact email so people can get in touch with you so you can click those check boxes when you've done those steps if you want to you're on boarded you have the downloaded textbook you're making time to participate in different ways what is actually going to be the participation there's many ways to learn and people are coming from a lot of different backgrounds part of what our group is um about oh sorry i just for the recording like wasn't fully sharing my own screen but here are the learning practices that we've just sketched and we can add more too so when you feel like contributing or paying attention to this textbook group here are some practices that might be interesting for you to check out and also comment on or improve like what are you doing when you're working with a textbook and that's going to range from peripheral reading and viewing could be experiential

it could be computational um especially as we move into the second half of the textbook where there's a lot more like model building there's also some uh templates which you can also in coda um like copy out to your own space if you'd like and there's some templates that are around tracking your practices in terms of your allocation of time and like keeping different um accounting of your engagement or your commitment there's no rubrics this can this is an aside document that only you have um other things you can bring in there uh but that's just to say you can use some practices and accounting if you like just how you see fit um and then the live meetings themselves you can see the live meetings from cohort one because they also were under the same open sourced with attribution approach and then here after this meeting this will be a link that is rewatchable and future meetings will be recorded as well but you can also see some discussions from earlier cohorts so that's a lot of the like logistics and onboarding we can see it goes through november that's the rough timing it's variable time commitment of different types through november strongly influenced by like what your availability is and what you're coming to and where you want to go and all of that what happens along the way are there any like logistical questions or advice or things that were missed all right um jessica and then anyone else and then maybe like alex or something can mention the overview um yes it's gonna fail so i'm jessica i'm also a returning um student participant for the course um and i guess i just wanted to add like um something that was really um really useful last time around was engaging in the discord with the different participants and kind of like learning from each other so also like the live informal chats you know and kind of asking questions of each other in on real time and to learn so those were like really helpful uh in addition to the ones you know asking the questions in like you know in the quota and and going over them here but it was also like that extra level of interaction that i really enjoyed last time so i just wanted to add that part thank you thanks jessica i added the discord link in the chat if anybody is not in it okay any other like just logistical comments does anyone want to mention a practice pattern that was relevant or they did that they didn't or didn't expect before we look at the pages where a lot of our shared work for people who choose to engage in the reading and writing this way before we even jump into any of it it's just anyone have an overview comment about a practice or a way that they felt like they were learning a lot for me the translation of the math into english was like mind-blowing um and actually like it wasn't so much the translation as like how much i learned in the process and how much i learned about like what i wasn't really understanding so um i think that that that was a practice that was like super instrumental to helping random material yeah and it's not

dumb like not even close so all right so in the next um you know 35 minutes there's three big areas to look at that are all part of the same function of us learning and collaborating they're also distinct though there's going to be some tables which are used as reference in coda so in coda where you can edit at is able to call like various types of things it can reference a page at which point that becomes a link to the page or it can link like what we'll explore in just a few seconds it can link a term like if you want to talk about variational free energy and then that on mouse over can be used to look at a lot of things like um the definitions and a lot of other aspects that we'll continue to have built so what is it um that these kinds of references are often too first let's look at the tables and then into the descriptions of the textbook and then the way that the discourse is range so it's not about like any importance or precedence it's just going through it in this order the ontology is a project that has been scaffolded at the institute since the beginning which is the development of term lists as well as their translations into multiple um languages so here's like a public face this link on the ontology page this is a public-facing version where definitions and then also the translation into these current languages so if anyone has a language also that they are not seeing here that they're familiar with that is there's 74 core terms and it can be an awesome contribution these ontology terms allow us to do a lot of things like again encoder referencing when we're using a term specifically which is going to facilitate like cross-linking as well as translation and a lot of other aspects um so we'll see how the ontology comes into play in the in the sections above that are more about the book but these are kind of static tables for semi-static tables that will be used in a lot of other parts above the live streams are a table of live streams the live stream series itself is paper driven and so if you're curious about a given topic like communication you may find that there are several live streams and relevant papers there inactive inference research we have a table of some citations and this is work from some knowledge engineering projects that are also always in progress in an area for people to be involved in with ways of aggregating resources bringing their accessibility into working documents like this and then there's just a list that's referenced elsewhere so those are kind of the static tables which are just some live streams that people may enjoy or are often brought up to contextualize other papers that we might mention and then the ontology table the ontology structures a lot of the technical discussions in the active field first the textbook and then the discourse that we'll have more on the textbook this section textbook is the textbook is open source hey gsrp do you have a question hi no i just i just managed to connect right now i'm not very familiar with the interface cool welcome thank you um the textbook section itself is going to contain many

tables and the table is going to be of the type that is described in the page so you can peruse these but they're used for multiple reasons first it's all enabled because the textbook is open source what the um kind of aliquoting of the textbook into these formats which is a lot of the work that was done in the first iteration of the first part for the first five chapters then we'll do that for the second having a field that's callable as like equation 2.1 means that anywhere you can just do at equation 2.1 and then okay okay that's what 2.1 looks and so these can also be tagged and these are contributions that are incredibly valuable for people to make if they feel moved to want to do these types of contributions so that like when people type in a term like bayesian a bunch of equations that are relevant come up then um blue maybe do you want to describe um what these descriptions are of the equations if you're there descriptions of you or ali do you want to describe sorry i was muted and i'm driving um okay okay i can uh describe really quickly they are um the descriptions are like literally how you would read the math equation so if you like are you know you read like y equals mx plus b this is like you know the um y -intercept is equal to the slope times the x -intercept plus the um or sorry the y -coordinate equals the slope times the x -coordinate plus the y or y -intercept right so it's like literally how you would read the equations in active inference using like english words for the terms that they represent thank you blue and those terms because they're referencing terms in the active inference ontology this is going to help contextualize the equation it's not the whole story on equations there's a ton of discourse below and then when we get to questions in our discourse you can ask open questions at reference or are stimulated by seeing formulas so as hopefully becomes clear all math backgrounds and preferences are welcome and everybody should be able to find uh meaning again for the people who feel like doing these kinds of annotations or work like for example jakub who showed some derivations and it was oh the first one goes to the second and the first one goes to the third but the second one doesn't go to the third well that's a different understanding even without the equation of how all of these terms which are natural everyday terms are linked to the formalisms because the formalisms are infinite journey with math and everything so that's why we're always looking to increase the accessibility of what is being related so that in the natural language use and in the settings where we're wanting to apply active inference we can interact with these things grounded in the formalisms but using regular ways of um communicating it's worth it because the equations are going to be peppered in the book as well as technical and difficult concepts or big questions that aren't equations so the difficulty is not purely analytical it's also just um everybody will have areas they're like familiar with and unfamiliar with okay figures is very

similar the figure information on the figure and then the image is posted so that figure 2.2 again it can always be clicked so especially as these tables gain completeness and also we can even copy the whole text of the book to github or to um pages so we can make and this is all on the road of developing interfaces for learning and applying active inference boxes and tables again similar to figures it just has images and it allows you to call box 2.1 okay that was what box 2.1 was code so this is going to come into play especially for people who have this background or would like to learn more and also in the second which is about model building and here we'll be developing i know ali and brock and jakob many people are interested in this kind of notebook driven development so again if somebody is familiar with like git and notebook development and hopefully there will be many opportunities there um in the chapters these are just uh this is a filtered view by chapter of the questions asked regarding that chapter if somebody wants to really focus on one section and also these are editable pages um or comment or if they're not where you can just add just anything you want on the chapters these are the sections of the book there's ten chapters and there's three appendices chapters one through five constitute like an epistemic learning component of the of the book and then chapters six through ten about building models and about applying still perhaps more theoretical than some would imagine applying to mean but relative to the first five it begins to get applied so these are just sections of the book there are the equations in a way that they're going to be referenceable in the discourse same for the figures boxes and tables and code is something that people can be developing more especially in the second part of the all right now to the discourse itself and the collapsing in the coda can start helping a lot just to go over these pages so a really important page especially for the live meetings is the questions so this questions page is going to be all the questions that have been asked however you can also go into cohort 2 questions to develop your own cohort specific so it's all referenceable like the underlying questions are all retained so if you don't want to like have your priors updated by what previous people have asked then don't look however if you want to see and even improve the kinds of questions and and the way that the live meetings work is that the questions that are for that week's topic we will look at like we'll go okay chapter one we're discussing today this one has the most up votes and then we'll just try to go through that list we're not going to address every possible one to the possible most possible especially written level of detail however before any time including after it's been discussed one time people can continue to come back and give thoughts and discourse one thing about the book structurally is there aren't any questions or cases provided so there's a large opportunity to develop really quality and

interesting questions from the very very basics of a chapter's gist or context or anything to really just open-ended questions where there isn't even an answer per se just kind of people can just work on some unsolvable problem or solve a problem or something just at the end of chapter one because kind of why not so we go through the interesting ones and clicking on this kind of an icon in a coded table will open up that where it's easier instead of having a text box that just becomes out of control this is like allowing a lot of discussion people are bringing in different resources we can ask questions about using terms in the ontology all these kinds of things again the questions can be um and also the questions can um reference parts of the book so oh this is about chapter one and figure 1.2 so that is how we'll do the live meetings adding as many questions as you would like is great just truly if whether it's something you think is very simple and you just want add it as something that somebody could be asked or something that is like okay just checking what is the difference between this and that that's a great or it can be open-ended or it can be anything that you're asking and then we'll just try to discuss them and continue to develop the discourse on them so that is because so just it's worth spending a lot of time on because the question structure the live meetings which is just the tip of the iceberg hopefully you would get a lot just from one hour per week of the discussions and seeing how people have updated the coda and their own notes however arguably you'll be developing a lot more if you're also asking questions and writing them down to yourself or at least adding some here so that's the questions aspect um ideas and insights don't have to be framed as questions but people can just add anything that they were thinking or reflecting on just like different ideas and insights from the book if they'd like um then we have several other aspects so next is the math learning group which was a great um innovation by several the participants here who are really active they know who they are and they all like helped steward and template a really accessible and inclusive approach to math discussions and this is an area for people on any side of some perceived math comprehension situation they can be in this group and we'll set sometimes amongst the people who want to commit to a certain set of times but it does it doesn't have to be recorded it's it was not previously and here are just several pages within the math learning group which is really everyone but that was kind of like the joke first there are several overviews written that are there was one table update so here if you see a strange sign these can be fixed it's just like okay you just have to replace it with that word but it was a previous table is being referenced but there's a few little cases like that math overviews were can be collaboratively improved to be like so they introduced section 2.2 this equation because it's going to show this just try to walk through some of you have overviews

of the book with the technical eye there are math learning resources that are just describing some of the resources people have found relevant for learning a lot of the background materials which have been named differently in different um settings and there's also a lot of change even in the fundamentals of active inference and free energy principle recently it's not like an absolute or total claim about what the bases or backgrounds are that being said certain areas like linear algebra and stochastic processes and statistics can be helpful anyone else can add more of course but i'm just mentioning a few areas and what the appendices are mentioning notation here is connecting some of the natural language ontology terms to the notation used in discrete time active inference and continuous time active inference so that we can then kind of keep some of the formalisms in mind and get familiar with some of the as we're talking about affordances or observations because that is the structure of the relationship of the ontology also not in its final form but at least there is a specific thing here's some math learning questions so it's just a separate section for questions if people don't want to put them in the main table but of course math questions again of the most confirmatory to the most difficult are also welcome in the questions table but these are some questions about conversations to have around learning and approaching the math there'll be some experimentation with um math typewriting i'll um mention the other sections before project ideas resources so here's some resources that people have found are as relevant also searchable here are some uh here's a feedback form which we'll update later in the cohort and a link for people who want to join a future cohort and then here's just ideation on what you would tell future participants different written feedback different ideas people have had about how to be involved in structuring the textbook group differently and better which should and will happen in the coming cohorts and then errata is uh just like some random stuff random thoughts people have and a few little typos or uncertainties in the book those kinds of things so just to kind of return to the overview and then we can have the rest of the time with any like discussions or thoughts people are having we are finishing our onboarding for this cohort 2 and then we're going to take two weeks per chapter and we're going to go through the first five chapters the chapters are not extremely extremely long in terms of page but there will be paragraphs where they're inscrutable or like you could also think about them for a long time that's where the practices come into play to whatever extent is possible having at least read through if or or listened if you prefer audio form to the chapter would be the best before a session on that chapter like ideally these are not motivating you to then check out chapter one it's the most helpful to have read the section before chapter one is a very overview so it will

spur many questions and it's non-technical and then all the equations and so on start beginning in chapter two try to paste things so that you are prepared for the discussion at least having read or listened to those pages and just looked at what the figures were and ideally also having contributed some questions or and even some discourse on the questions or thoughts on it so that in the live meeting it's not like okay here we've read the question who has a thought on it it could be like well here's two perspectives people have shared is there any other thought on how that could happen so well in chapter one which again is very overview styled these will be pretty open discussions about the context of active inference maybe even ongoing changes in active one overall note is this mod this textbook there's of course many ways to think about it but it's like describing the kernel a framework which is the action perception loop and some formalisms and approaches towards formalisms for the action perception loop the papers conferences literature at a very rapid pace is elaborating in a ton of directions which actually i will leave to ali to describe because i think you'll have a lot to say all this well yeah uh actually on resources page uh i've accurate curated a collection of 17 papers uh and i've put them roughly in the order of increasing technicalities i think these papers might come in handy as we go through the textbook because i mean the textbook is especially some chapters are really dense and will definitely need some additional resources feel free to add any other papers or articles articles you came across or reorder them in any way you see fit but yeah this is uh this page might be useful uh for some people if they want to gain a little bit more deeper understanding in terms of the essential concepts and so on thanks yes jessica um yeah i guess i wanted to build up on at least um comment because i found like there were a lot of like the papers that he uh suggested that were very helpful like i'm a pretty much beginner and in this space and also with a lot of the math and and a lot of the papers that he added and which was sort of like straddling between so like statistics and also um what the book and is talking about on is it i think just i guess to like back up a little bit like one of the challenges that i found is like in the textbook it's frame a lot of the statistical material and some of the math is framed from the active inference obviously and approach and the link back to some of the basics in statistics might be lost and there were like some references that suggested and mentioned that allowed me to be able to make the connections to understand like okay so inactive inference one is looking at this term that's normally used in statistic in this other way this is how you know it connects and and be able to learn a little bit of the foundation for that and then be able to um make the transition to how it's used in an active inference and framework or more than this approach and so yeah so that's what i wanted to like build up because they were like you

know very like useful on connection to kind of go from more foundational aspect to then connecting it to active inference and how they are applied in this sense thank you jessica yeah i i think it's great that the math group opened up a lot of that discussion in space because this is a textbook and to kind of return to that idea of it being a this is going to be focusing on a didactic slash exemplar single agent case one entity involved in an action perception loop and it's going to describe that process it also doesn't describe the um every aspect from every possible perspective it's a short book overall and the step by step which is probably linked in all these the step-by-step guide for example is also almost book length you'll find other almost book length approaches to learning including ones that take very different approaches like the path that they take in a step-by-step guide with even some of the same authors is not the same as the textbook and the textbook structure is not without uncertainties and so on there's like from the fundamental mathematical to especially chapter five when the neuroscience is going to get connected broader discussions around systems and complexity and cybernetics control theory philosophy and especially the introduction of the 4e etc embedded extended things like nested communicating cyber physical systems are not going to be in the middle of chapter two except potentially by just mention the research has a huge amount of these cases and like ramifications of active which often you'll see with an adjective before like deep active inference now it's considering something about temporal depth or something so again it's all say the textbook is a lot and yet there's more especially in application so that again just returns us to the practices and thinking about this as learning the kernel of active inference or a kernel or a distilled representation by some of the people who have been working in this area for a long time like carl fristen and those who are bringing a sort of view that and physics-based view which is mainly explored in recent literature but also not in the anyone could have more thoughts on that or just general thoughts on the practices or way to approach the textbook even though published in march or april this year still it feels like a lot has changed in inactive inference or what do people think what is the way to approach the ali and then anyone else well uh just one thing that i think um might be helpful to keep in mind is that is that um as uh if people want to go through the the additional resources and the other papers in the literature uh well uh it would be helpful to keep in mind that the notations used in the is mostly different from most of the notations used in the rest of the literature so uh this might be confusing at the initial stages so um yeah that's one caveat that i wanted to mention thanks for mentioning it um it's one reason why we have this language ontology driven approach because we'll be able to make columns for different papers this is written down elsewhere i just want to note that for

people who like do want to really parse through a paper where there is alignment in what the papers are discussing or or disagreement or ambiguity or all the above but is about the terms and the concepts which is also what connects the formalisms whether the math analytical equation formalisms or the computational coding implementations whatever formalisms are being used when they're being discussed outside of formalisms it's with these kinds of terms so decision making and policy selection affordances capacities for action and then that's represented a certain way in equations maybe as a bold letter or maybe as a vector or there's just that is where heterogeneity is the ways that certain um terms are represented because people are coming at this for many threads which we'll be learning about but in um statistics maybe they're used to talking about like y is a predictor of x but then in bayesian models of perception and cognition maybe o is used for observation and s is for a hidden state that might be an analogous situation actual topics and how they're related yeah any other thoughts on this oh the last section which of here is the project ideas jessica or anyone want to give an overview on the project ideas um yes um i guess like in general the idea is if you have on a specific way that you want to practice like learning by doing on active inference or like a project in mind and to share it and with the community to see if there's all the other people who are interested in joining you and exploring it in that um manner and and so yeah so you share the idea um you put in your name and so that other people can contact you on and also tap themselves on saying that they're interested and and then you know you can also like organize yourself and to create like a separate um study group or like you know project and group and to do this um together and then there's already like some um projects idea that already like emotion and that you can also learn by doing so like black friends and which is also part like the institute itself and it's a project but then they're all the ones that are also like more independent from all the previous um participants in the you know study group on yeah so it's very very informal and just you know post a flag and and see you know what momentum generates and you know what feedback you get out of it thank you jessica yeah there's just people can continue to add and it looks like some people even are adding things but it ranges from audio recordings of the book going into various aspects of the rigor and accessibility of the textbook uh actin foundations course which i know brock and ali have worked on and other types of textbook enrichment um people interested in domain specific cases many of which we can scaffold really well in active block friends all the meetings this meeting is in gather so we can be a little bit more like face to face also potentially use breakouts spatially but also be not related to the discord if people don't want to be that way it's

not that it always maybe we can you know see how things evolve in platforms but we're here so that we can have a really focal study and connection time and the project meetings are in our those or or they can be arranged as people see fit but by default in the discord like for active block for instance for others and that helps provide some visibility for your project even if someone's not in the textbook group and um as the months go by hopefully people see a lot of ways to want to apply active inference the book will make several comments to that effect too does anyone have any other comments otherwise we'll end this first discussion neil and then anyone else yeah i mean this all looks absolutely fantastic i was just wondering what is it that acting uh what's the objective for the acting group what are you trying get out of this and what might we help to achieve that well appreciate that i hope we can find out and co-create how in general our mission at the institute is to address education and research so those are the two sections of the and these types of groups and people's engagement in them and the projects other facilitations of the institute as they see relevance in a changing ecosystem they're all um rowan uh yes i just wanted it's a very simple in regards to the questions for the upcoming chapters do we are we able to put those questions in before the meeting is that is that the aim so as i'm reading the chapter and i've got a question i can pop it in there and then that'll be answered during the session yes so you could just like click in which chapter you want the question to be in and then just hit enter dot dot okay new question dot dot and then you can um add if you want in this field you can use the at and say okay at equation 2.1 and then the question could be like i saw equation one dot dot but yes adding them before will make those discussions really um rich and have some starting places for the discussion and then if someone can't make the discussion that's also fine because then their questions will be addressed and yeah be given visibility so we can continue to address it good thank you all right any final comments you